Factor Exposure Analysis

Extension to the eight-ETF portfolio study · daily returns 2021-2025 · independent project

Method

Daily portfolio excess returns (net of the one-month T-bill rate) are regressed on the Fama-French market, size (SMB), value (HML), profitability (RMW), investment (CMA) and momentum factors from the Kenneth R. French Data Library. OLS with an intercept, Newey-West standard errors with 5 lags, alphas annualised with 252 sessions. The active return Main minus Benchmark is regressed on the same factors without the risk-free rate. US factors are applied to all ETFs, including international ones (see limitations).

Findings

• Main's US market beta is 0.55 versus 0.61 for the benchmark, although both hold 60% equity: the international sleeve moves less than one-for-one with the US market (VEA 0.77, VWO 0.61). The active market beta is -0.054 (t -11.5).

• The active return also loads on size (SMB +0.063, t +11.0), value (HML +0.030, t +3.6) and, negatively, on profitability (RMW -0.031, t -4.3). These factors explain 37% of daily active-return variation; the rest is likely non-US equity and currency exposure, which US factors do not span.

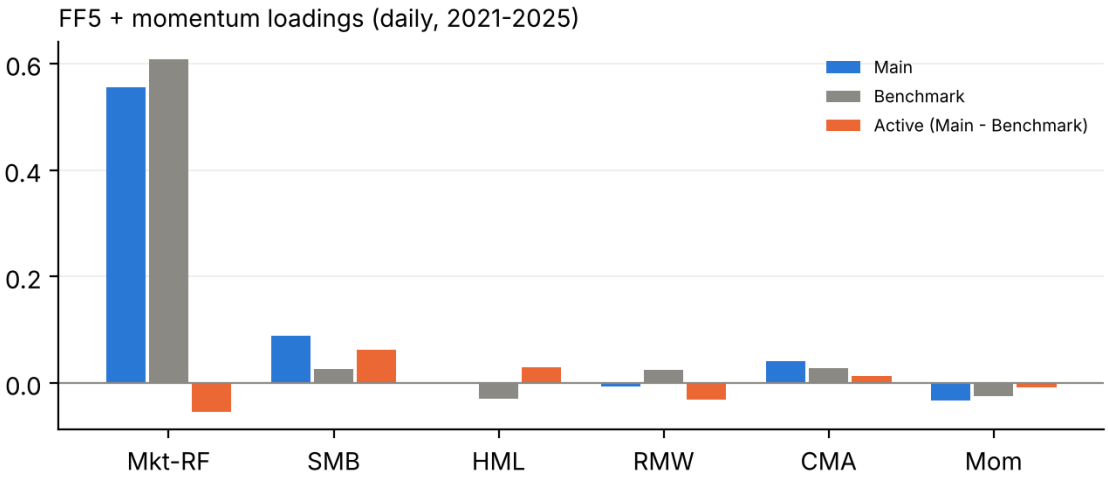
• Average active return was -1.48% a year (arithmetic). The lower market beta cost -0.58% in a strong US market, the small-cap tilt -0.27% and profitability -0.16%, while value added +0.22%. The remaining alpha of -0.67% has t = -0.9: factor tilts account for about 55% of the shortfall, and the rest cannot be distinguished from noise.

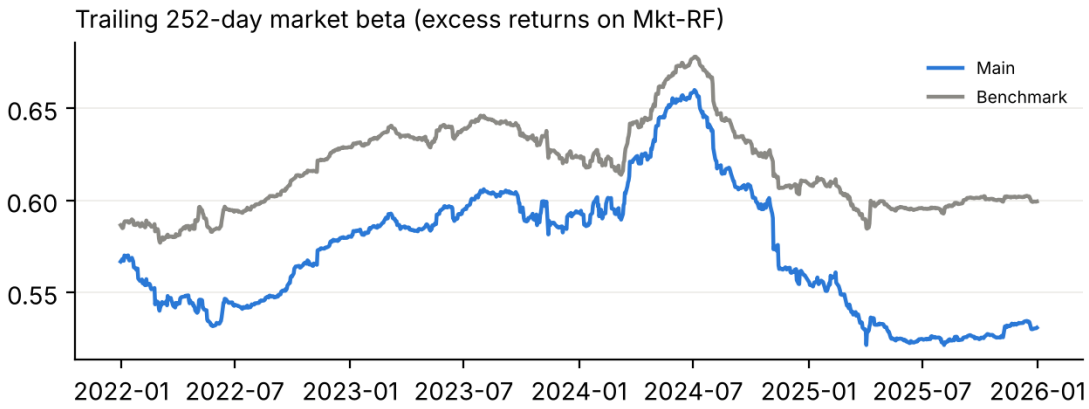
• US equity factors describe the bond ETFs poorly (R2 BND 0.09, IEF 0.06, BNDX 0.06) and the international equity ETFs only partly (VEA 0.70, VWO 0.48). Portfolio-level alphas therefore absorb unmodelled bond and foreign returns and are not interpreted; the active-return regression, where bonds largely cancel, is the informative one.

• Adding RMW, CMA and momentum to the three-factor model raises the active-return R2 only from 0.35 to 0.37; size and market explain most of what the US factors can explain.

Loadings (FF5 + momentum)

Series	Alpha/yr	t	Mkt-RF	SMB	HML	RMW	CMA	Mom	R2
Main	-2.19%	-1.5	+0.55	+0.09	0.00	-0.01	+0.04	-0.03	0.91
Benchmark	-1.52%	-1.5	+0.61	+0.03	-0.03	+0.03	+0.03	-0.03	0.95
Equal	-2.60%	-1.5	+0.44	+0.11	-0.01	-0.02	+0.05	-0.03	0.81
Conservative	-2.50%	-1.5	+0.37	+0.07	-0.03	0.00	+0.04	-0.04	0.75
VTI	-0.31%	-1.1	+1.00	0.00	+0.01	+0.02	+0.03	-0.01	1.00
VB	-2.97%	-1.9	+1.02	+0.59	+0.14	-0.06	+0.02	-0.03	0.97
VEA	-2.03%	-0.5	+0.77	+0.02	+0.14	-0.06	+0.07	-0.05	0.70
VWO	-2.23%	-0.4	+0.61	+0.05	-0.03	-0.22	+0.10	-0.02	0.48
BND	-3.01%	-1.3	+0.04	+0.06	-0.09	+0.04	+0.04	-0.05	0.09
IEF	-3.58%	-1.1	0.00	+0.08	-0.13	+0.06	+0.06	-0.06	0.06
TIP	-2.36%	-1.0	+0.06	+0.02	-0.04	0.00	+0.06	-0.03	0.05
BNDX	-2.99%	-1.4	+0.02	+0.05	-0.07	+0.04	+0.03	-0.03	0.06
Active	-0.67%	-0.9	-0.05	+0.06	+0.03	-0.03	+0.01	-0.01	0.37





Active return decomposition (annualised)

Component	Contribution
alpha	-0.67%
Mkt-RF	-0.58%
SMB	-0.27%
HML	+0.22%
RMW	-0.16%
CMA	+0.01%
Mom	-0.04%
total (mean return x 252)	-1.48%

Verification and limitations

Coefficients match numpy least squares (max difference 9.9e-16); the decomposition sums to the mean active return (difference 3.5e-18). Factor files: F-F_Research_Data_5_Factors_2x3_daily_CSV.zip, F-F_Momentum_Factor_daily_CSV.zip, downloaded 2026-10-15, SHA-256 recorded. Limitations: US factors for international ETFs understate their explanatory power (lower R2 is expected for VEA, VWO, BNDX); bond exposures are not captured by equity factors; daily close timing differs for foreign markets; five years is a short sample; factor loadings describe, they do not prove cause.

Source: Kenneth R. French Data Library, https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html